NOTES:

There are some errors in my code (I couldn't score separately for the second wall and I couldn't create random '1's on the first wall, and I didn't put a '2' in the second wall because I had already passed the first and second walls by collecting 32 points) BUT MY CODE IS WORKING.

Game Design and Objective

This game allows a student to progress by collecting first and second-grade courses and reaching the final target. The objective is for the player to collect a certain number of course symbols (1 and 2) and gather enough ECTS points to pass the first and second years. At the end of the game, the total ECTS points and the total number of moves are displayed.

Game Board

The game board is 16x16 in size. On the board, there are symbols representing the student (P), course symbols (1 and 2), the target (X), and walls (#). The walls surround the first and second-grade courses. The student is positioned at the exact center of the board.

Code Workflow

- The `initializeGame()` function initializes the game board and randomly places course symbols.

- The `printBoard()` function prints the game board to the screen.

- The `movePlayer(char direction)` function controls the player's movement and updates the game board accordingly.

- The main function `main()` initiates the game, takes input from the user, checks the end of the game, and displays appropriate messages.

Code Explanation

- `#define` statements define certain values as constants. These constants include the board size, wall width, required ECTS points, and symbol definitions.

```
void initializeGame() {
    // This function initializes the game by setting up the game board.
    // It randomly places second-grade course symbols on the board,
    // places first-grade course symbols within a 5x5 square area in the center,
    // and places the target symbol (X) at the bottom right corner.
}

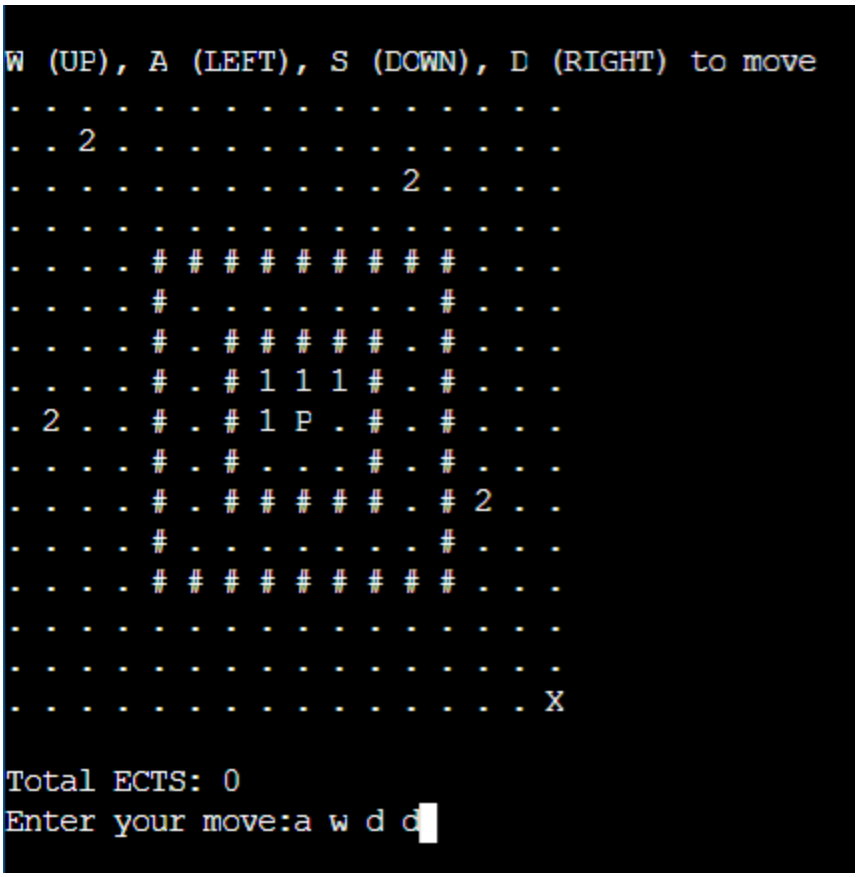
void printBoard() {
    // This function prints the current state of the game board to the console.
    // It iterates over each cell in the board, printing the corresponding symbol.
}

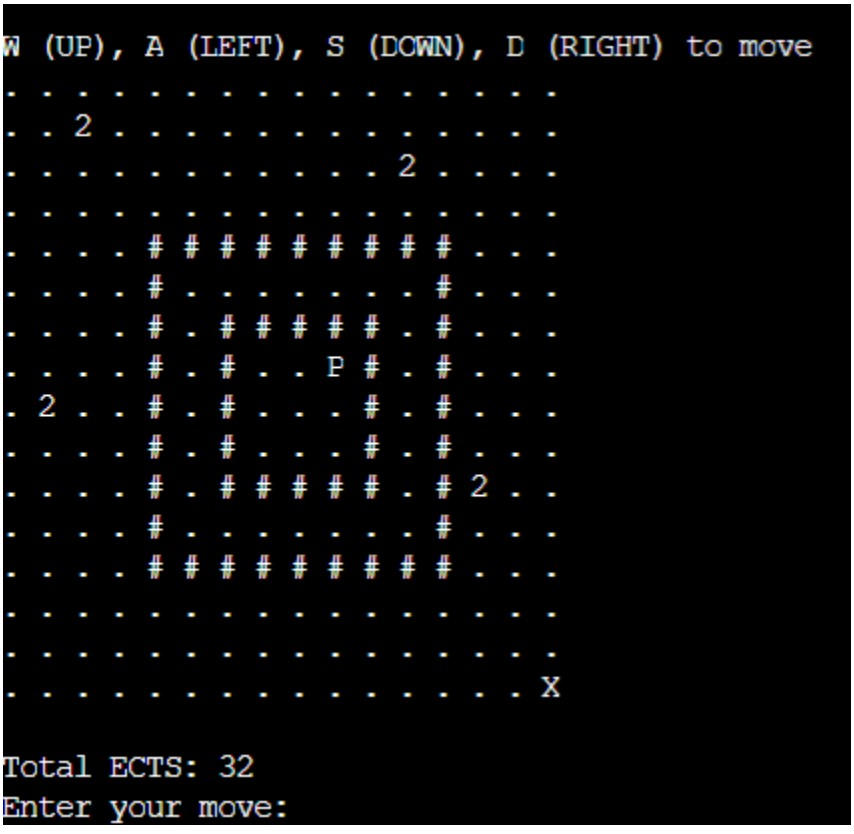
void movePlayer(char direction) {
    // This function handles the movement of the player based on the input direction.
    // It calculates the new position of the player and updates the game board accordingly.
    // It also checks for collisions with walls, course symbols, and the target,
    // and performs appropriate actions based on the game rules.
}
```

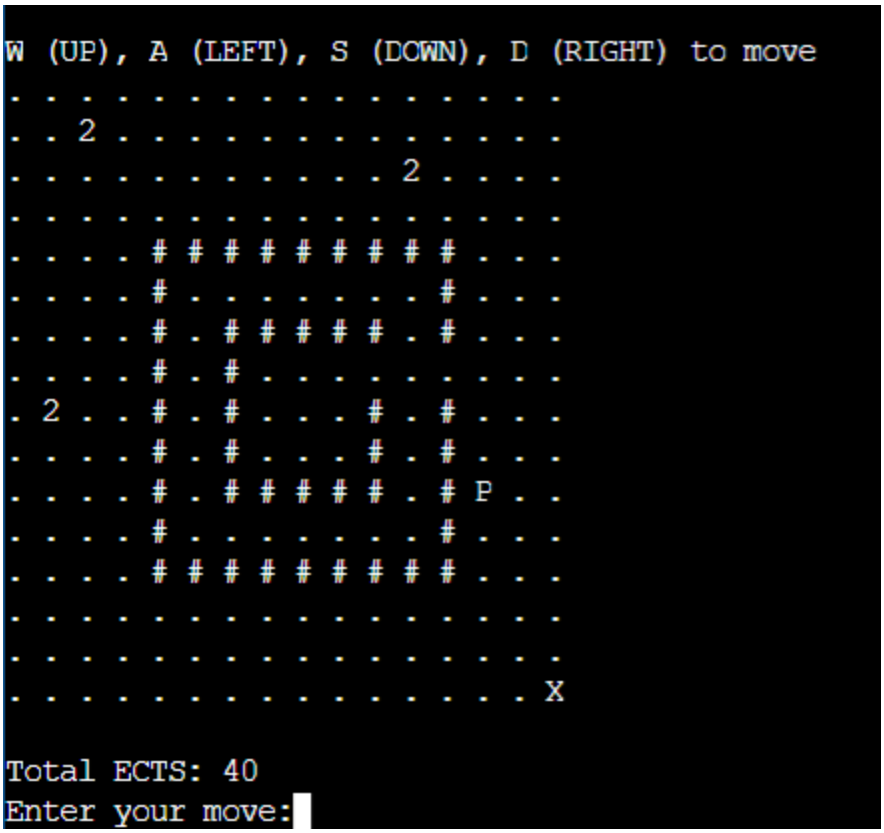
Conclusion

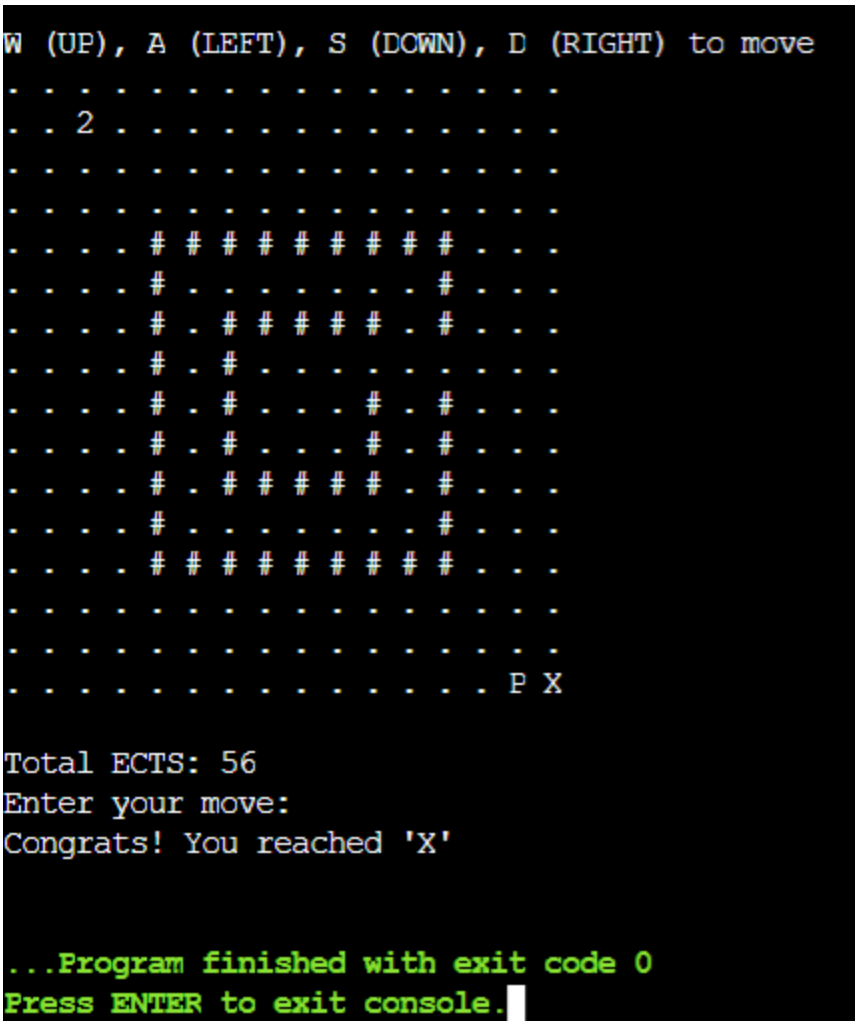
This game simulates a student's progression by collecting courses and reaching the final target. The objective is to collect courses and earn enough ECTS points to progress. The game requires careful planning of moves and strategy development from the player.

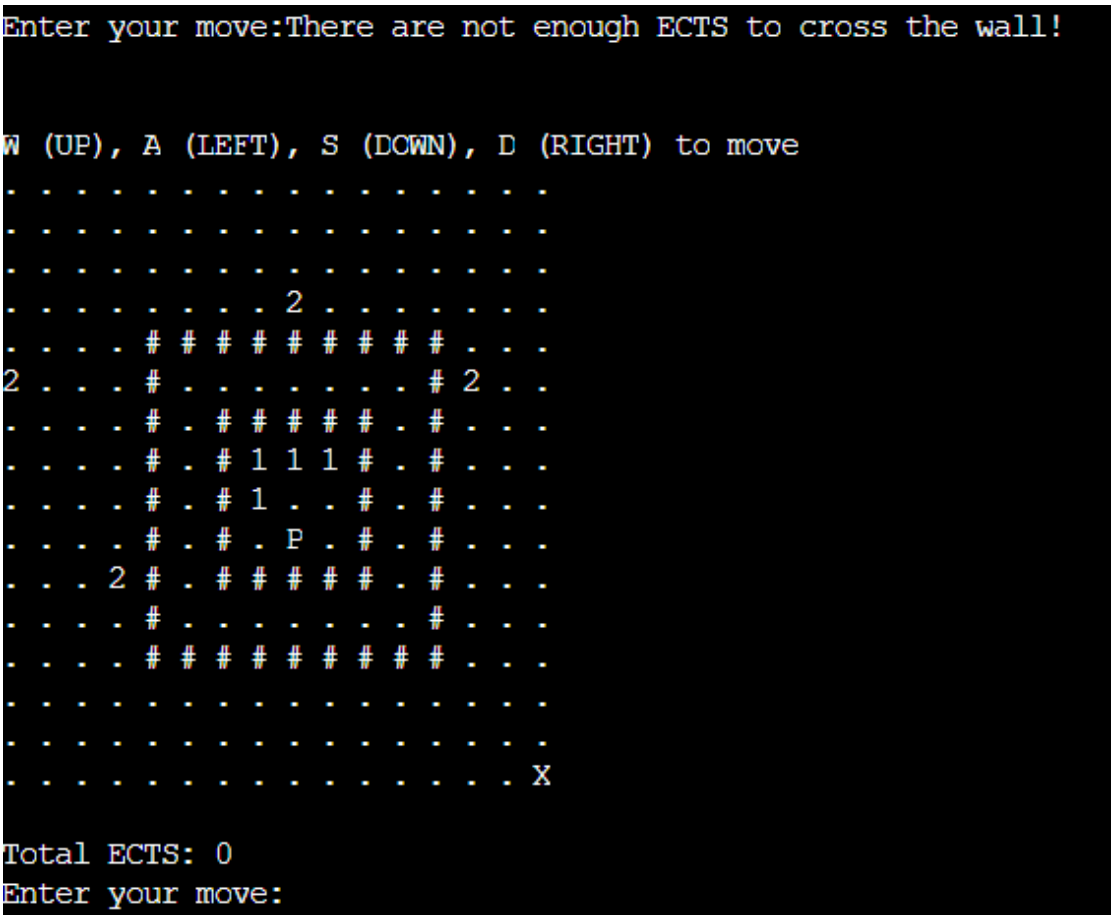
OUTPUTS:

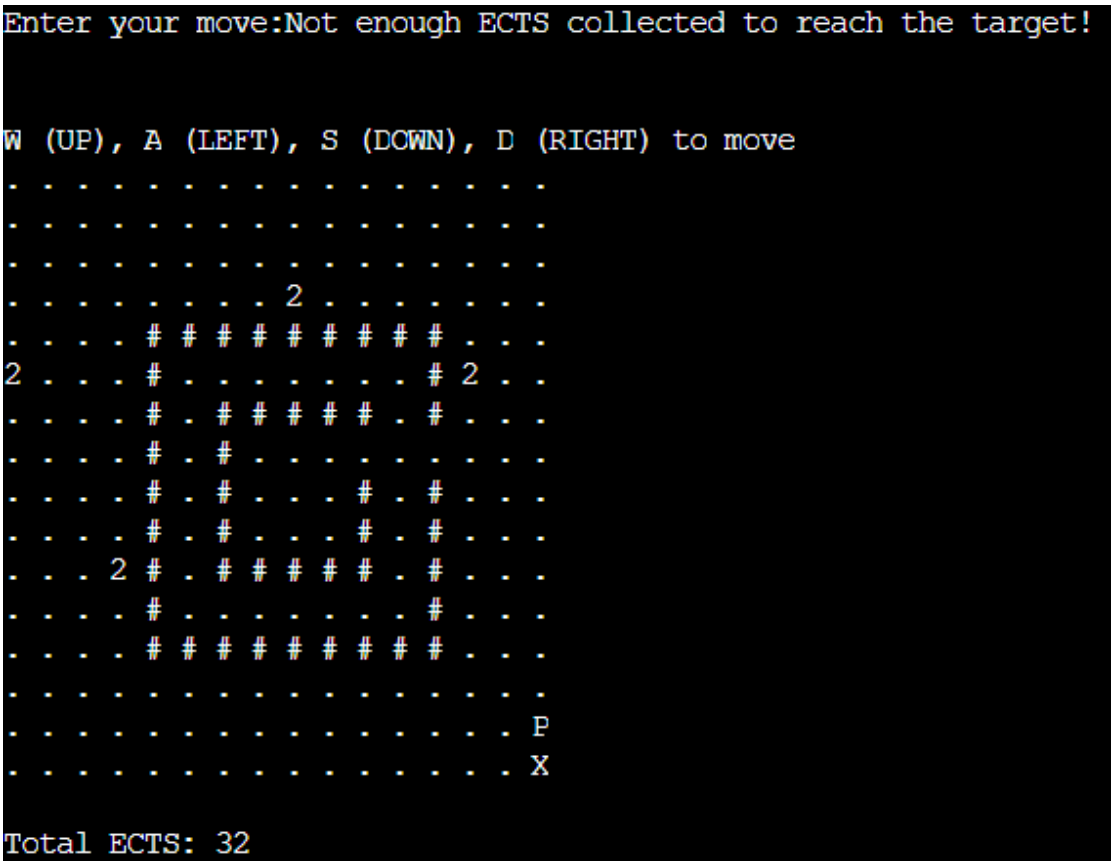

W (UP), A (LEFT), S (DOWN), D (RIGHT) to move
Total ECTS: 0
Enter your move:a w d d


W (UP), A (LEFT), S (DOWN), D (RIGHT) to move
Total ECTS: 32
Enter your move:


W (UP), A (LEFT), S (DOWN), D (RIGHT) to move
Total ECTS: 40
Enter your move:


W (UP), A (LEFT), S (DOWN), D (RIGHT) to move
. . . . . .
. . 2 . . . . . .
. . . . . .
. . . . . .
. . . . # # # # # # # # # . . .
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. . . . . .
. . . . P X
Total ECTS: 56
Enter your move:
Congrats! You reached 'X'
...Program finished with exit code 0
Press ENTER to exit console.


Enter your move:There are not enough ECTS to cross the wall!
W (UP), A (LEFT), S (DOWN), D (RIGHT) to move
. . . . . 2 . . . . . .
. . . . # # # # # # # # # . . .
2 . . . # . . . . # 2 . .
. . . . # . # # # # # . # . . .
. . . . # . # 1 1 1 # . # . . .
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. . . . # . . . . # . . .
. . . . # # # # # # # # # . . .
. . . .
. . . .
. . . . X
Total ECTS: 0
Enter your move:


Enter your move:Not enough ECTS collected to reach the target!
W (UP), A (LEFT), S (DOWN), D (RIGHT) to move
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. . . .
. . . . 2 . . . .
. . . . # # # # # # # # # . . . .
2 . . . # . . . . # 2 . . .
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. . . . # . . . . # . . . .
. . . . # # # # # # # # # . . . .
. . . .
. . . . P
. . . . X
Total ECTS: 32